PART 9

From mapped reads to a consensus genome

Build an N-masked consensus FASTA from read evidence.

ONE REFERENCE. TWO EVIDENCE-BASED DECISIONS.

REFERENCE GENOME: 2,400 BASES

↓

WHAT THE READS SHOW

One supported base change and a region with no reads

CONSENSUS FASTA (SCHEMATIC)



TEAL DOT: BASE CHANGE AT 877

Reference is G; the mapped reads support A.

YELLOW: 350-BASE COVERAGE GAP

No reads at 1,051–1,400, so the bases are masked as N.

Without read evidence, we cannot confirm a nucleotide at that site.

The exercise also masks any other positions with depth below 10x.

CONCEPT

What is a consensus sequence?

One representative sequence built from aligned reads.

REFERENCE

The sequence we map reads against

READS

Observed bases and confidence at each position

CONSENSUS

A decision at each position, or N when evidence is insufficient

KEY IDEA

Consensus is a summary, not a raw read and not a complete picture of mixed populations.

NUCLEOTIDE EVIDENCE

When reads disagree with the reference

We inspect depth, base quality and allele support.

SCHEMATIC ALIGNMENT NEAR POSITION 877



Reference = G | Majority read base = A | Candidate SNP

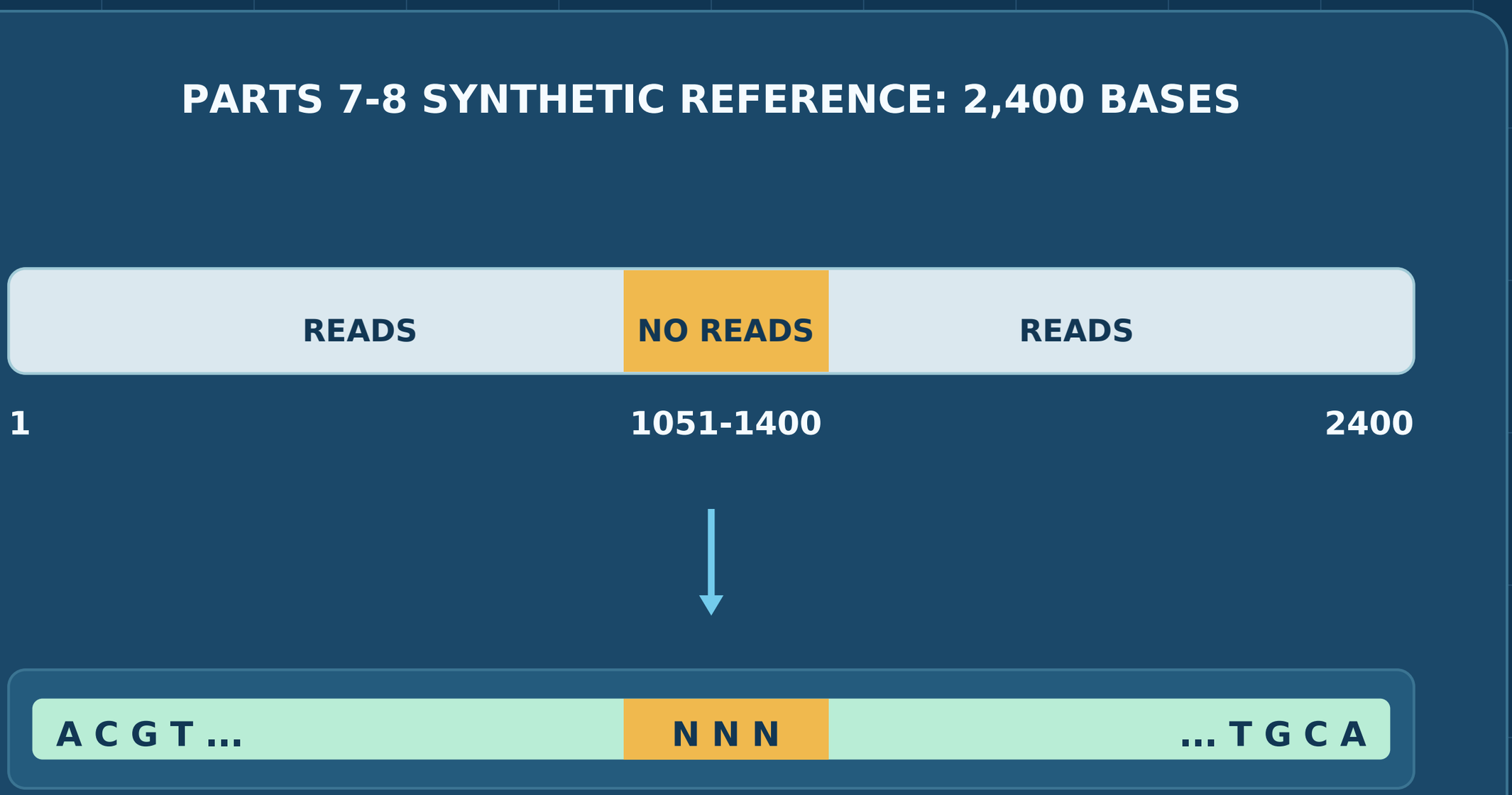
DO NOT ASSUME

A mismatch alone is not enough. Real calls require suitable depth, quality, mapping and strand evidence.

LOW COVERAGE

What if no reads cover a position?

Do not silently copy the reference base into a reported genome.



THRESHOLD

We also mask positions below 10x for this exercise. That is a teaching choice, not a universal cutoff.

PRACTICE DATA

Same reference. One new synthetic SNP.

We retain the Part 8 gap but update the reads for variant calling.

reference.fasta

The same 2,400-base reference from Parts 7 and 8.

reads_part9.fastq.gz

300 synthetic reads with G to A at position 877.

350-base zero-depth gap remains at positions 1051-1400.
Do not use the old Part 7/8 FASTQ for the new SNP exercise.

SYNTHETIC DATA

The files are artificial training data. They are not real patient, virus or clinical sequencing results.

SET UP + MAPPING

Install the final tool and map the reads

Continue working in Ubuntu/WSL2.

1. Install BCFtools (one time only)

```
sudo apt install bcftools -y
```

2. Open the Part 9 working folder

```
mkdir -p ~/bioinformatics_training/part9
cd ~/bioinformatics_training/part9
explorer.exe .
```

3. Copy both input files into the folder

```
ls -lh reference.fasta reads_part9.fastq.gz
```

4. Align reads and create an indexed BAM

```
minimap2 -ax sr reference.fasta reads_part9.fastq.gz > mapped.sam
samtools sort -o mapped.sorted.bam mapped.sam
samtools index mapped.sorted.bam
```

VARIANT CALLING

Which bases differ from the reference?

BCFtools examines the aligned reads and writes candidate variants.

1. Index the FASTA, then create genotype likelihoods

```
samtools faidx reference.fasta
bcftools mpileup -Ou -f reference.fasta mapped.sorted.bam > likelihoods.bcf
```

2. Call candidate variants (haploid teaching example)

```
bcftools call -mv --ploidy 1 -Oz -o variants.vcf.gz likelihoods.bcf
```

3. Index the VCF

```
bcftools index variants.vcf.gz
```

SYNTHETIC DESIGN TARGET: position 877, G to A
Inspect the actual VCF before treating a variant as confirmed.

REMEMBER

A variant call is evidence-based, not simply a mismatch seen in one read.

VARIANT FILTERING

Not every candidate is trustworthy

A basic quality filter is a starting point, not a clinical SOP.

1. Retain calls with site QUAL at least 30

```
bcftools filter -i 'QUAL>=30' -Oz -o passing.vcf.gz variants.vcf.gz
```

2. Index the filtered VCF

```
bcftools index passing.vcf.gz
```

BASE QUALITY is confidence in an individual base call.

VARIANT QUAL is a quality measure for a site-level variant call.

They are related, but not the same measurement.

IMPORTANT

Mixed infections, minor alleles, strand bias and sequencing artifacts require more careful analysis.

LOW-DEPTH MASK

Turn the depth table into a BED mask

A BED file marks positions that should become N.

1. Measure depth at every reference position

```
samtools depth -aa mapped.sorted.bam > depth.tsv
```

2. Write BED intervals for all positions below 10x

```
awk -v OFS='\t' '$3<10 {print $1,$2-1,$2}' depth.tsv > lowcov.bed
```

DEPTH FILE
Position 1051
1-based

BED INTERVAL
Start 1050, end 1051
0-based, end-exclusive

The zero-depth gap 1051-1400 is masked through these BED intervals.

DETAIL

The AWK output contains one 1-base interval per low-depth position. Adjacent intervals may be merged later.

MAKE THE FASTA

Create the masked consensus

Apply passing variants and replace low-depth positions with N.

Run this single command:

```
bcftools consensus -f reference.fasta -m lowcov.bed passing.vcf.gz > consensus.fa
```

POSITION 877

A replaces reference G if the call passes.

POSITIONS 1051-1400

N marks the deliberately uncovered gap.

CRITICAL DISTINCTION

At unmasked sites with no alternate call, BCFtools keeps the reference base.

That does NOT prove the base was directly observed.

WET-LAB LINK

Insufficient evidence should be reported as uncertainty, not invented sequence.

CHECK YOUR OUTPUT

Inspect before sharing the consensus

Look at the variant record, sequence length and masked positions.

1. Inspect the passing VCF records

```
bcftools view -H passing.vcf.gz | head
```

2. Index your FASTA

```
samtools faidx consensus.fa
```

3. Check consensus length and count Ns

```
grep -v '^>' consensus.fa | tr -d '\n' | wc -c
grep -v '^>' consensus.fa | tr -cd 'N' | wc -c
```

DESIGN EXPECTATIONS: 2,400-base reference; one SNP at 877; 350-base known gap plus other positions below 10x.

Actual outputs depend on the mapper and variant caller.

REPRODUCIBILITY

Record your depth cutoff, filtering criteria and software versions.

10-MINUTE CHALLENGE

Your turn: explain the consensus

The goal is interpretation, not just a FASTA file.

1. Why can a mapped read still leave gaps in genome coverage?
2. What evidence supports the introduced G to A change?
3. Why do we convert low-depth reference positions to N?
4. Does a single consensus capture a mixed viral population?

NEXT: PART 10

Multiple-sequence alignment. How do we compare consensus genomes?